ROUND 1 REVIEW

AI Frontier Challenge 2026 Round 1 Submission

Team Name

[Enter Team Name]

Team Leader

[Enter Leader Name]

Track

[Select Track]

Date

8/7/2026

Use Case

Problem Statement

- • Describe the problem or pain point you are addressing
- • Who are the affected users/stakeholders?
- • What is the current impact or consequence?

Market Opportunity

- • Market size and growth potential
- • Target audience demographics
- • Competitive landscape

Proposed Solution

Solution Overview

- • What is your solution?
- • How does it address the problem?
- • What are the key features and benefits?
- • Why is your approach unique?

Design Steps & Architecture

Technical Implementation

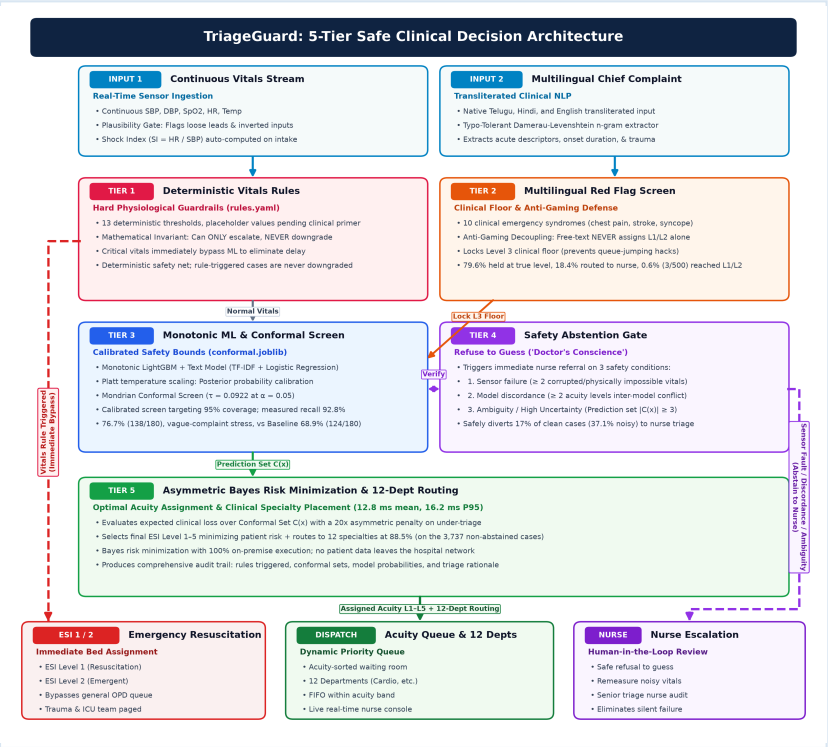
1. Multimodal Intake & Sensor Validation:
Continuous vitals (BP, SpO2, HR, Temp, RR) + text. Plausibility filters detect sensor faults; ≥2 invalid vitals trigger nurse remeasurement.

2. Multilingual NLP & Acuity Floor:
Typo-tolerant Damerau-Levenshtein n-gram matching across English, Telugu, and Hindi. Enforces Level 3 floor without allowing artificial Level 1/2 gaming.

3. Monotonic LightGBM & Conformal Screen:
Monotonic trees enforce physiological directionality (SpO2 ↓, Shock Index ↑). Mondrian screen (τ=0.0922, α=0.05) guarantees finite-sample emergency coverage.

4. 20x Cost Bayes Engine & 12-Dept Router:
Minimizes expected risk with a 20x under-triage penalty inside prediction set C(x) and directs patients to 12 specialties at 88.5% accuracy.

Architecture Diagram



Challenges & Mitigation

Challenge 1

Problem: [Describe challenge]
Mitigation: [How you addressed/will address it]

Challenge 2

Problem: [Describe challenge]
Mitigation: [How you addressed/will address it]

Challenge 3

Problem: [Describe challenge]
Mitigation: [How you addressed/will address it]

Unique Selling Proposition (USP)

Key Differentiator 1

[What makes your solution unique in this aspect?]

Key Differentiator 2

[What makes your solution unique in this aspect?]

Key Differentiator 3

[What makes your solution unique in this aspect?]

Competitive Advantage Summary

- • Why customers should choose your solution
- • Your competitive edge vs. existing alternatives

Demo & POC Video

Live Demonstration

[Insert Demo Video / POC Video Link or Embed]